

5-2-1: Examining Hardware-Based Network Tools

At the end of this episode, I will be able to:

1. Identify and explain common hardware-based network tools, given a scenario.

Learner Objective: *Identify and explain common hardware-based network tools, given a scenario*

Description: In this episode, the learner will examine common hardware-based network troubleshooting tools. We will explore tools such as tone generators, cable testers, taps, and more.

- What are some of the common cabling tools?
 - o Cable cutter
 - ♣ A tool used to cut through various types of cables, including electrical wires and optical fibers, for precise cut for installations or maintenance work.
 - o Cable stripper
 - ♣ A tool designed to remove the insulation from electrical wires and optical fibers without damaging the conductor or fiber inside.
 - o Cable crimper
 - ♣ A tool used to join connector ends to various types of cables, including:
 - ♣ Coaxial
 - ♣ Ethernet
 - ♣ Fiber optics
 - o Punchdown tool
 - ♣ A tool used to insert and secure wires into punchdown blocks or keystone modules.
 - How can we ensure the cables are working properly?
 - o Cable Tester
 - ♣ Verifies the electrical connections in a signal cable or other wired assembly, testing for continuity, resistance, and ensuring no shorts or open circuits.
 - o Visual Fault Locator
 - ♣ Emits a red laser light to visually identify breaks, bends, or faults in fiber optic cables, aiding in troubleshooting and maintenance.
 - o Optical Time-Domain Reflectometers (OTDRs)
 - ♣ are diagnostic tools used to characterize optical fibers by measuring light reflections and losses, identifying faults, and determining fiber length and splice locations.
 - What if you need to locating cables in a build?
 - o Tone generator and locator
 - ♣ A device used to trace and identify wires or cables in a bundle, ensuring proper routing and connectivity.
 - How about monitoring data on a cable?
 - o Network tap
 - ♣ A tool that allow monitoring of network traffic transparently, enabling data capture without interrupting the flow.
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- Additional Resources:
 - o If applicable